

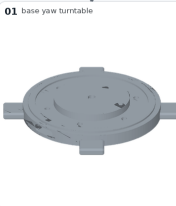
Robotic Arm STL Exploded Assembly Guide

Generated from the uploaded STL files. The exploded layout is inferred from part names and geometry, so it is best used as a visual assembly map. Final screw sizes, servo horns, bearings, and spacers should be confirmed on the real printed parts.

Exploded Assembly Sketch - Full 6-Axis Robotic Arm

Approximate assembly order inferred from the STL filenames and matching geometry. Left/right shell parts open outward in this sketch; close them around the joint axis and wiring during assembly.

1. Base yaw



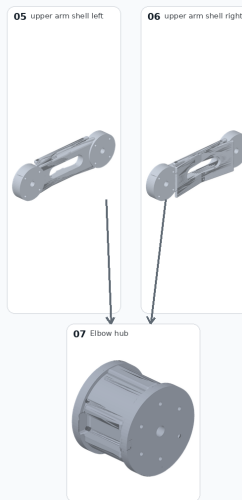
Column drops into/onto turntable. Confirm smooth rotation before adding arm load.

2. Shoulder frame



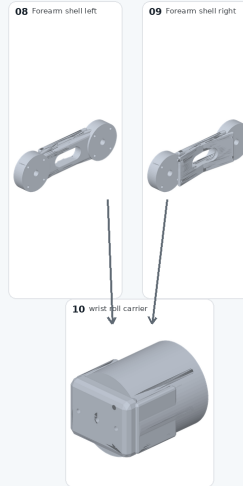
Mount left/right yokes on the rotating column, leaving the center space for the shoulder pivot.

3. Upper arm



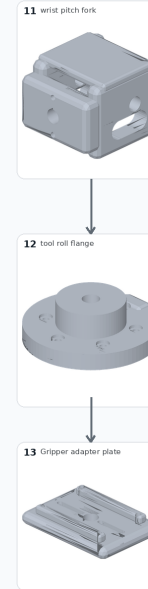
Close the two upper arm shells around the shoulder/elbow axis path, then add the elbow hub at the far end.

4. Forearm



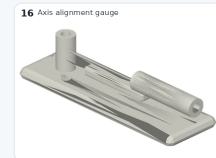
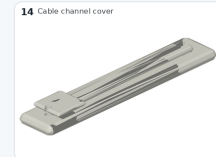
Attach forearm shell pair to elbow hub; wrist roll carrier fits at the distal end.

5. Wrist + tool



Pitch fork -> tool roll flange -> adapter plate -> gripper.

6. Covers / spacers / tool



Install spacers during joint assembly; cable cover goes on after cable routing. Gauge is for alignment only.

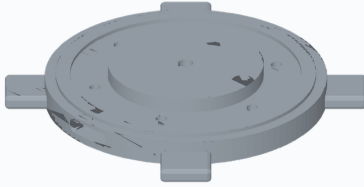
Final chain: 01 base -> 02 column -> 03/04 shoulder yokes -> 05/06 upper arm shells -> 07 elbow hub -> 08/09 forearm shells -> 10 wrist roll carrier -> 11 wrist pitch fork -> 12 tool flange -> 13 gripper adapter. 14 cover, 15 spacers, and 16 gauge are auxiliary/alignment items.

Part Index - Rendered from Actual STLs

Robotic Arm STL Parts - Individual Object Drawings

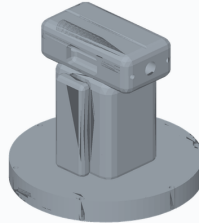
Each thumbnail is rendered directly from the uploaded STL geometry. Dimensions are bounding-box extents in millimeters if the STLs were modeled in mm.

01. J1 base yaw turntable
159.6 x 159.6 x 22.0 mm



Bottom fixed base / yaw turntable.

02. J1 rotating column
108.0 x 108.0 x 93.5 mm



Sits on/inside the base and carries the shoulder.

03. J2 shoulder yoke left
14.8 x 58.0 x 99.0 mm



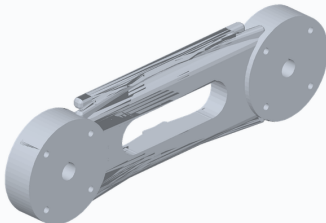
Left shoulder side plate; flat side faces inward.

04. J2 shoulder yoke right
14.8 x 58.0 x 99.0 mm



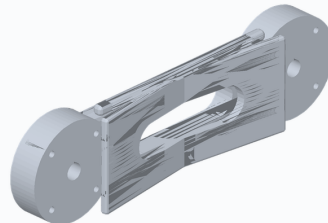
Right shoulder side plate; flat side faces inward.

05. J3 upper arm shell left
116.0 x 168.0 x 50.0 mm



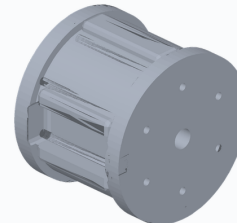
Left half of upper arm shell.

06. J3 upper arm shell right
116.0 x 168.0 x 50.0 mm



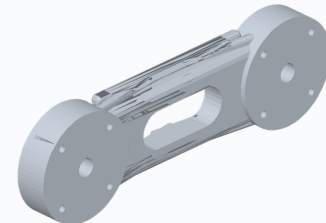
Right half of upper arm shell.

07. Elbow hub
55.0 x 68.0 x 68.0 mm



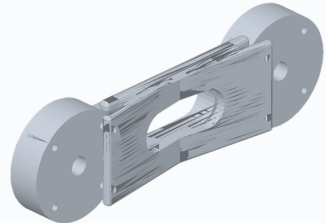
Joint hub between upper arm and forearm.

08. Forearm shell left
115.8 x 168.0 x 50.0 mm



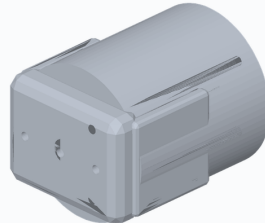
Left half of forearm shell.

09. Forearm shell right
115.8 x 168.0 x 50.0 mm



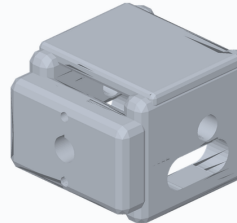
Right half of forearm shell.

10. J4 wrist roll carrier
54.0 x 67.5 x 54.0 mm



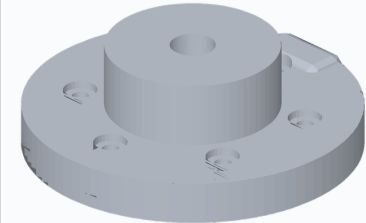
Wrist roll carrier at end of forearm.

11. J5 wrist pitch fork
49.0 x 56.0 x 43.5 mm



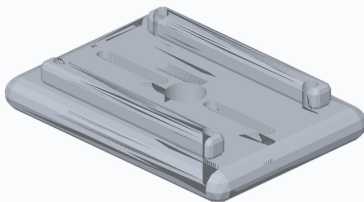
Fork part around the wrist pitch axis.

12. J6 tool roll flange
62.0 x 62.5 x 23.0 mm



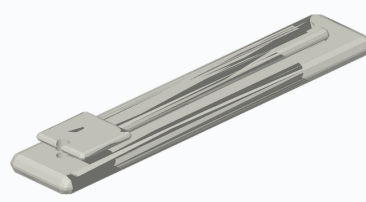
Tool roll / output flange.

13. Gripper adapter plate
62.0 x 44.0 x 11.5 mm



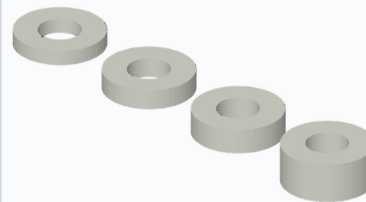
Interface plate for gripper.

14. Cable channel cover
20.0 x 104.0 x 8.5 mm



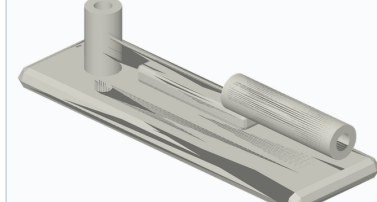
Cover for cable-routing channel; install after wires.

15. Spacer set
90.0 x 18.0 x 8.0 mm



Spacers/washers used on pivot axes as needed.

16. Axis alignment gauge
89.0 x 28.0 x 20.5 mm



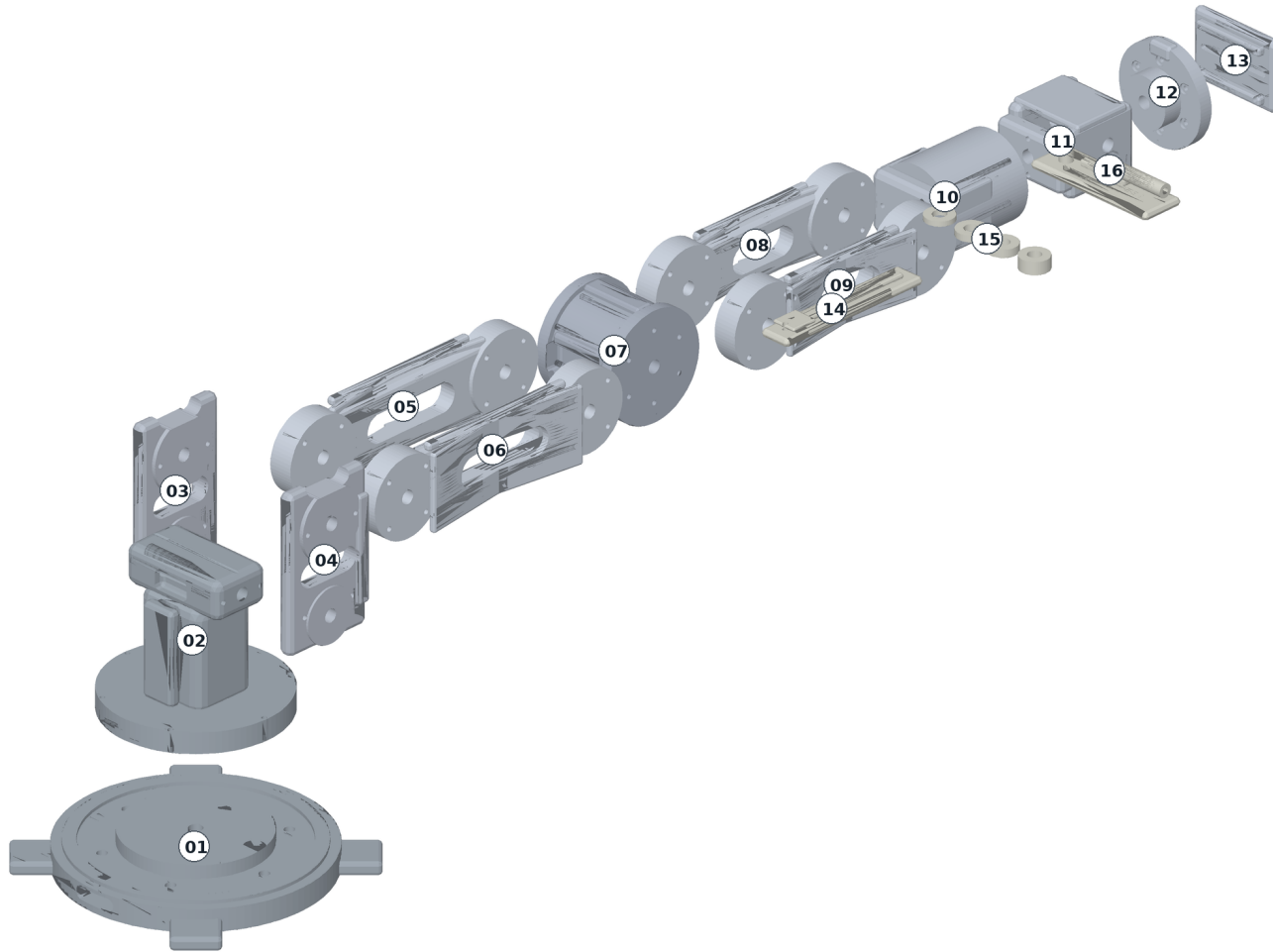
Assembly/alignment tool; not a permanent arm part.

Approximate 3D Exploded View

Number labels correspond to the STL filenames. Left/right shell pairs are shown separated so the assembly sequence is visible.

Exploded 3D View - Actual Uploaded STL Meshes

Approximate position/orientation only. Use hole alignment, servo horns, bearings/spacers, and screw fit to finalize physical assembly.



Assembly Order Summary

Step	Parts	What to do
1	01 + 02	Place the J1 rotating column into/on the base yaw turntable. Confirm smooth yaw rotation before adding the arm load.
2	03 + 04	Attach the left/right shoulder yokes to the rotating column. Keep the center gap aligned for the shoulder pivot axis. The flat inner faces should face each other.
3	05 + 06 + shoulder hardware	Bring the upper arm shell halves together around the shoulder joint path. Do not trap servo wires; route cables before final tightening.
4	07	Add the elbow hub at the distal end of the upper arm. Use spacer pieces where needed so the hub rotates without rubbing the shells.
5	08 + 09	Attach the forearm shell halves to the elbow hub, again leaving any cable path accessible until motion is checked.
6	10	Mount the wrist roll carrier at the distal end of the forearm.
7	11 -> 12 -> 13	Add the wrist pitch fork, then the tool roll flange, then the gripper adapter plate. The gripper mounts to the adapter plate.
8	14 + 15 + 16	Install the cable channel cover only after wiring is tested. Use the spacer set during pivot assembly. Use the alignment gauge as a tool; it is not a permanent arm part.

Important fit check: do a loose dry-fit first, move each joint through its range by hand, then tighten screws gradually. If a shell rubs a hub or yoke, add/remove one of the spacer pieces before final tightening.

Individual Part Drawings

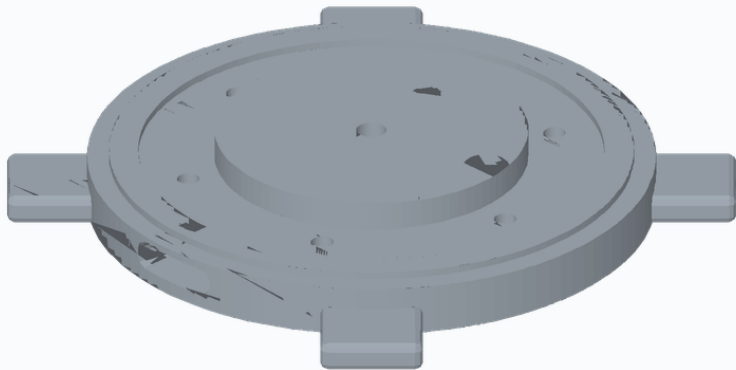
Each following page shows one STL object with an isometric drawing plus front, top, and side projection views. Bounding-box dimensions are from the STL coordinates, assumed to be millimeters.

01. J1 base yaw turntable

Bounding box: 159.6 x 159.6 x 22.0 mm | STL local bounds: X -79.8 to 79.8, Y -79.8 to 79.8, Z -7.0 to 15.0

Bottom fixed base / yaw turntable.

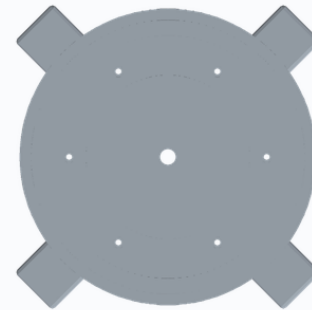
Isometric STL render



Front



Top



Side

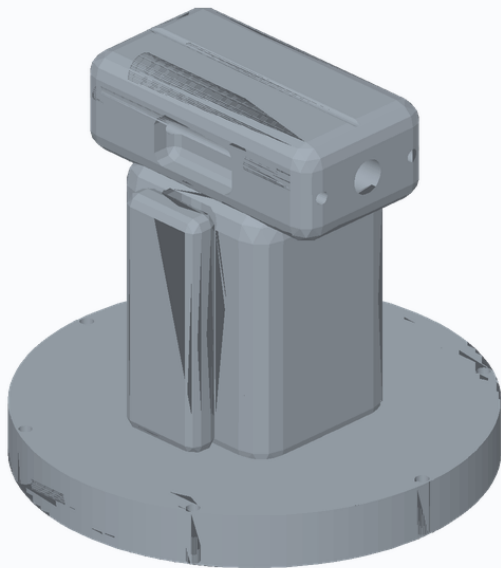


02. J1 rotating column

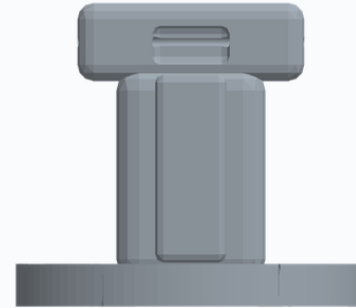
Bounding box: 108.0 x 108.0 x 93.5 mm | STL local bounds: X -54.0 to 54.0, Y -54.0 to 54.0, Z -6.5 to 87.0

Sits on/inside the base and carries the shoulder.

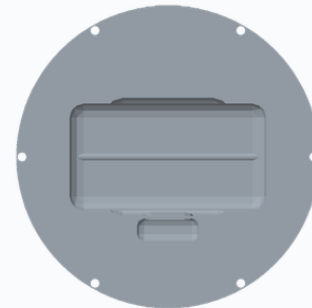
Isometric STL render



Front



Top



Side



03. J2 shoulder yoke left

Bounding box: 14.8 x 58.0 x 99.0 mm | STL local bounds: X -7.8 to 7.0, Y -29.0 to 29.0, Z -50.0 to 49.0

Left shoulder side plate; flat side faces inward.

Isometric STL render



Front



Top



Side



04. J2 shoulder yoke right

Bounding box: 14.8 x 58.0 x 99.0 mm | STL local bounds: X -7.0 to 7.8, Y -29.0 to 29.0, Z -50.0 to 49.0

Right shoulder side plate; flat side faces inward.

Isometric STL render



Front



Top



Side

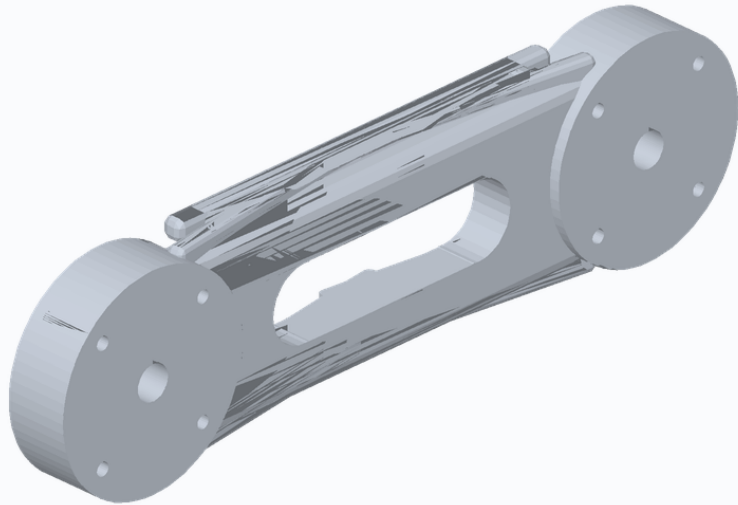


05. J3 upper arm shell left

Bounding box: 16.0 x 188.0 x 50.5 mm | STL local bounds: X -8.5 to 7.5, Y -94.0 to 94.0, Z -25.0 to 25.5

Left half of upper arm shell.

Isometric STL render



Front



Top



Side

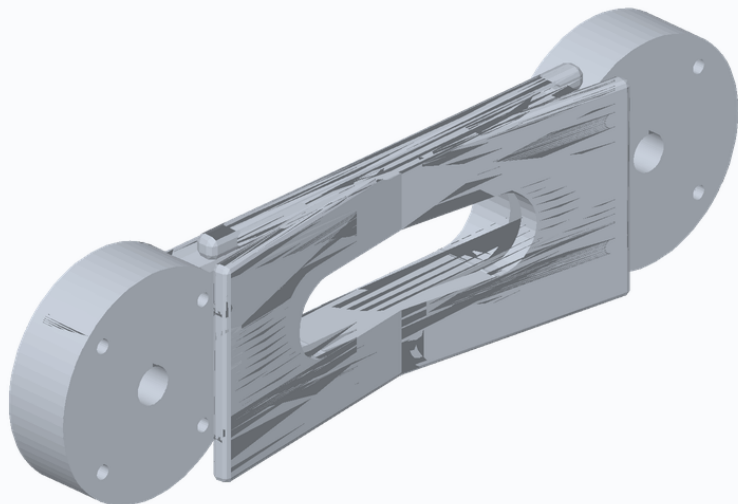


06. J3 upper arm shell right

Bounding box: 22.9 x 188.0 x 50.5 mm | STL local bounds: X -7.5 to 15.4, Y -94.0 to 94.0, Z -25.0 to 25.5

Right half of upper arm shell.

Isometric STL render



Front



Top



Side

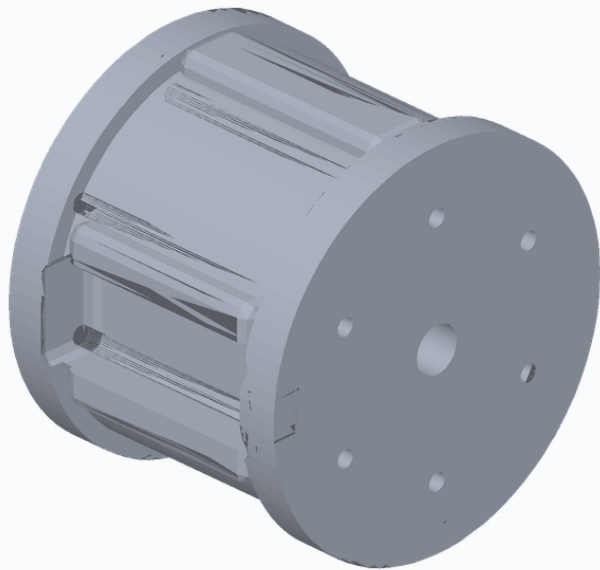


07. Elbow hub

Bounding box: 55.0 x 68.0 x 68.0 mm | STL local bounds: X -27.5 to 27.5, Y -34.0 to 34.0, Z -34.0 to 34.0

Joint hub between upper arm and forearm.

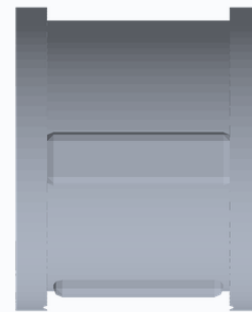
Isometric STL render



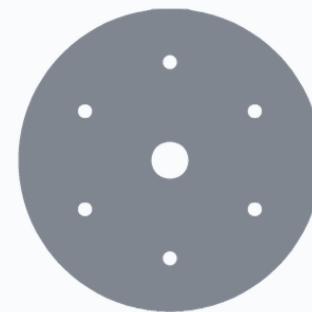
Front



Top



Side

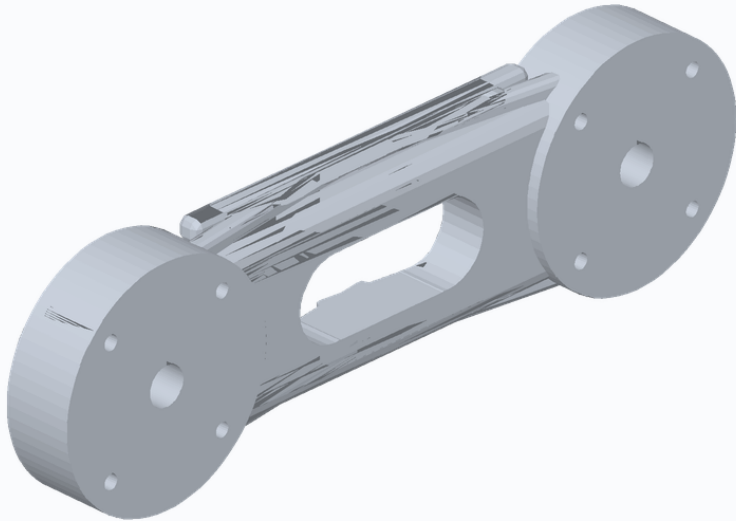


08. Forearm shell left

Bounding box: 15.8 x 168.0 x 50.0 mm | STL local bounds: X -8.3 to 7.5, Y -84.0 to 84.0, Z -25.0 to 25.0

Left half of forearm shell.

Isometric STL render



Front



Top



Side

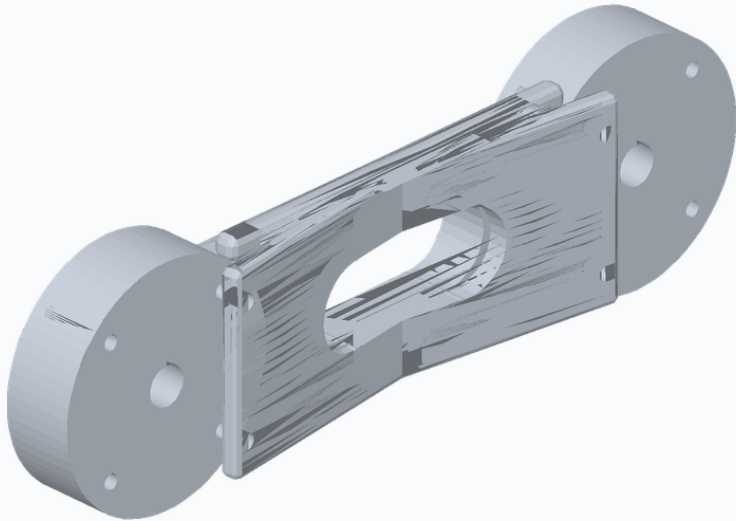


09. Forearm shell right

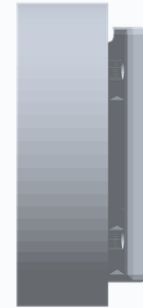
Bounding box: 21.8 x 168.0 x 50.0 mm | STL local bounds: X -7.5 to 14.3, Y -84.0 to 84.0, Z -25.0 to 25.0

Right half of forearm shell.

Isometric STL render



Front



Top



Side

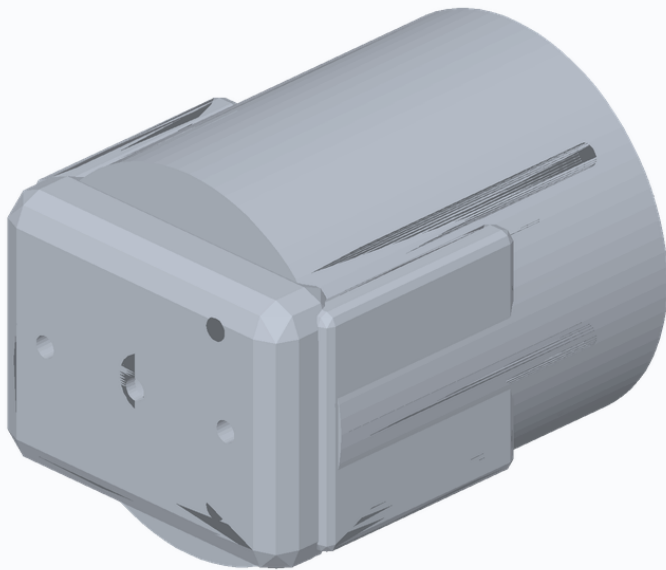


10. J4 wrist roll carrier

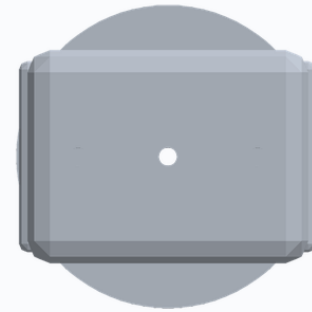
Bounding box: 54.0 x 67.5 x 54.0 mm | STL local bounds: X -27.0 to 27.0, Y -38.5 to 29.0, Z -27.0 to 27.0

Wrist roll carrier at end of forearm.

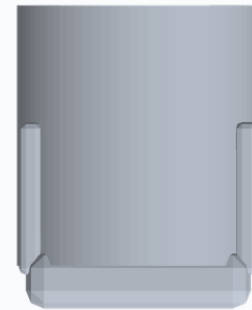
Isometric STL render



Front



Top



Side

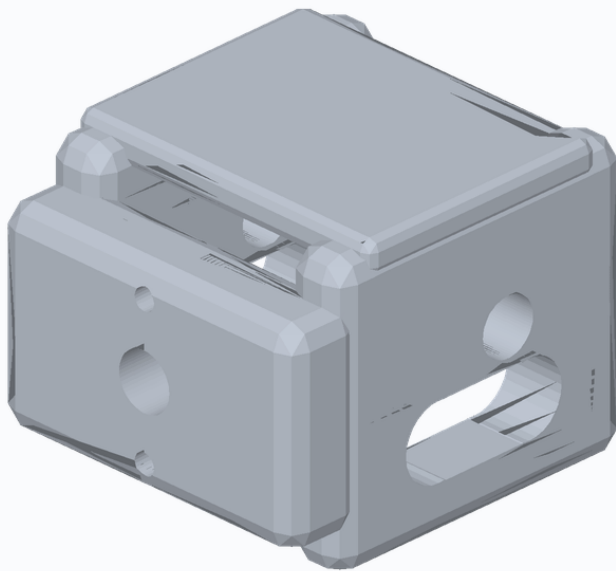


11. J5 wrist pitch fork

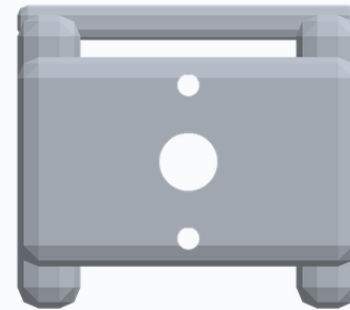
Bounding box: 49.0 x 56.0 x 43.5 mm | STL local bounds: X -24.5 to 24.5, Y -24.0 to 32.0, Z -21.0 to 22.5

Fork part around the wrist pitch axis.

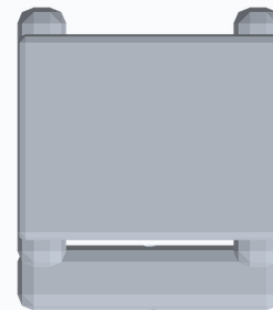
Isometric STL render



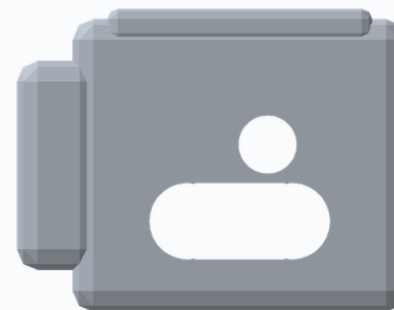
Front



Top



Side

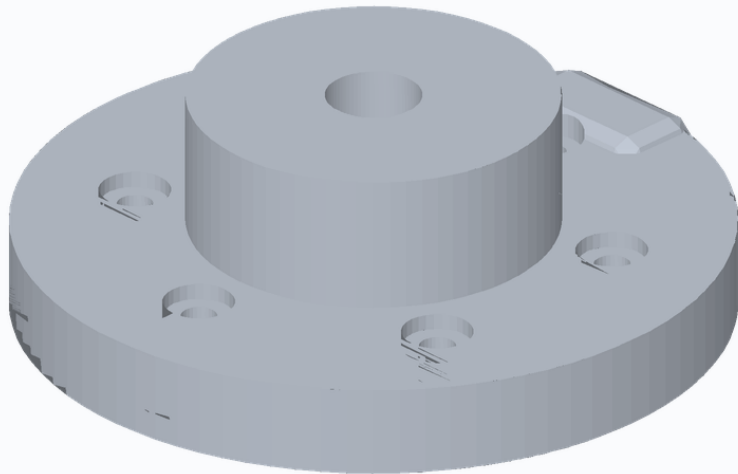


12. J6 tool roll flange

Bounding box: 62.0 x 62.5 x 23.0 mm | STL local bounds: X -31.0 to 31.0, Y -31.0 to 31.5, Z -7.0 to 16.0

Tool roll / output flange.

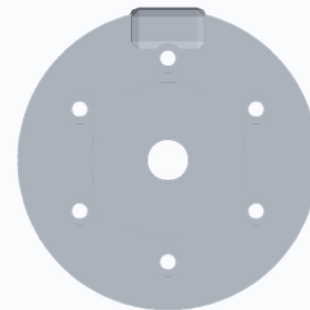
Isometric STL render



Front



Top



Side

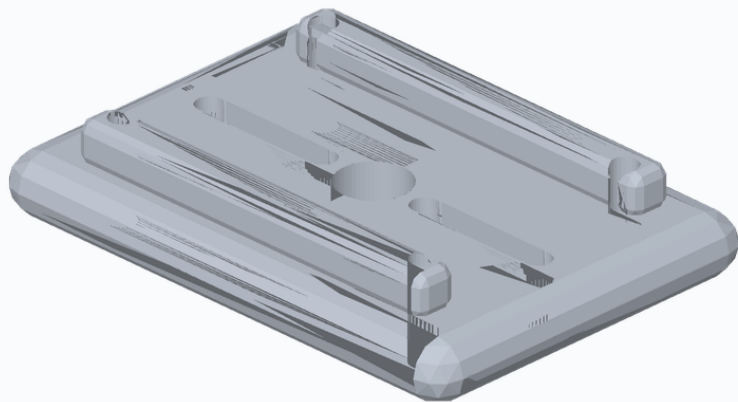


13. Gripper adapter plate

Bounding box: 62.0 x 44.0 x 11.5 mm | STL local bounds: X -31.0 to 31.0, Y -22.0 to 22.0, Z -3.5 to 8.0

Interface plate for gripper.

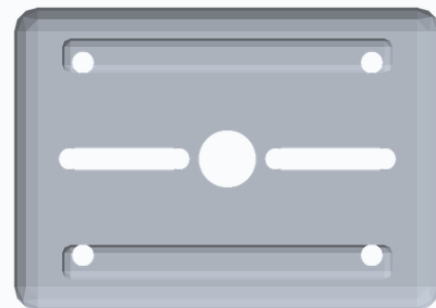
Isometric STL render



Front



Top



Side

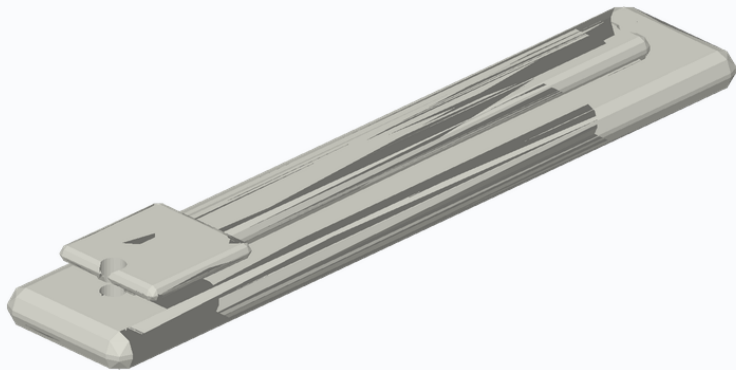


14. Cable channel cover

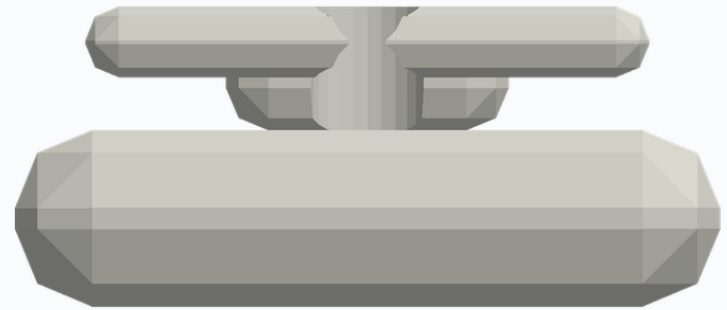
Bounding box: 20.0 x 104.0 x 8.5 mm | STL local bounds: X -10.0 to 10.0, Y -52.0 to 52.0, Z -2.5 to 6.0

Cover for cable-routing channel; install after wires.

Isometric STL render



Front



Top



Side

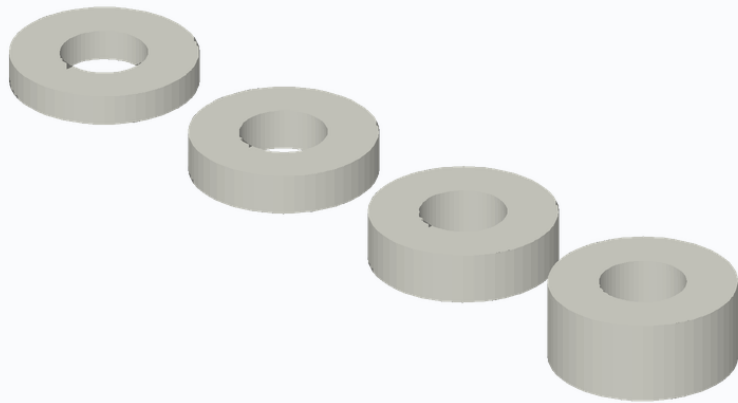


15. Spacer set

Bounding box: 90.0 x 18.0 x 8.0 mm | STL local bounds: X -45.0 to 45.0, Y -9.0 to 9.0, Z -4.0 to 4.0

Spacers/washers used on pivot axes as needed.

Isometric STL render



Front



Top



Side

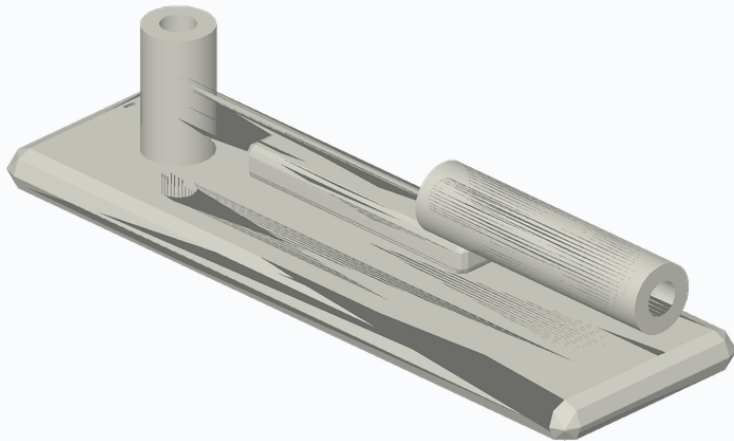


16. Axis alignment gauge

Bounding box: 89.0 x 28.0 x 20.5 mm | STL local bounds: X -44.0 to 45.0, Y -14.0 to 14.0, Z -2.5 to 18.0

Assembly/alignment tool; not a permanent arm part.

Isometric STL render



Front



Top



Side

